COMPUTATIONAL PERFORMANCE OF HYBRID ROCKET MOTOR LOADED WITH FUEL AND ADDITIVES

Capstone Project
Submitted in Fulfillment of the Requirement for Award of the Degree

Of

BACHELOR OF TECHNOLOGY

In

AEROSPACE ENGINEERING

By

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2025-26

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CERTIFICATE

I hereby certify that the work which is being presented in the capstone entitled **“Computational Performance of Hybrid Rocket Motor Loaded with Fuel and Additives”** in partial fulfillment of the requirement for the award of degree of Bachelor of Technology and submitted in Department of Aerospace Engineering, Lovely Professional University, Punjab is an authentic record of my own work carried out during period of Capstone under the supervision of Mr. Harikrishna Chavhan, Assistant Professor, Department of Aerospace Engineering, Lovely Professional University, Punjab.

The matter presented in this capstone has not been submitted by me anywhere for the award of any other degree or to any other institute.

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Jitesh Jambulkar,
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Talla Gowtham Kumar

This is to certify that the above statement made by the candidate is correct to best of my knowledge.

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Mr. Harikrishna Chavhan
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The B-Tech capstone examination of Jitesh Jambulkar, Laxmipriya Murmu, Mounish Damera, Talla Gowtham Kumar has been held on

Signature of Examiner

DECLARATION

I hereby declare that the research work reported in the dissertation proposal entitled "Computational Performance of Hybrid Rocket Motor Loaded with Fuel and Additives" in fulfilment of the requirement for the award of Degree for Bachelor of Technology in Aerospace Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under supervision of my research supervisor Mr. Harikrishna Chavhan. I have not submitted this work elsewhere for any degree or diploma.

I understand that the work presented herewith is in direct compliance with Lovely Professional University's Policy on plagiarism, intellectual property rights, and highest standards of moral and ethical conduct. Therefore, to the best of my knowledge, the content of this dissertation represents authentic and honest research effort conducted, in its entirety, by me. I am fully responsible for the contents of my dissertation work.

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TOPIC APPROVAL PERFORMA

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Supervisor Name : Harikrishna Chavhan **UID :** 28597 **Designation :** Assistant Professor

Qualification : ____________________ **Research Experience :** ____________________

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PROPOSED TOPIC : Oxidizer effect on combustion of propellant in hybrid rocket engine via computational analysis

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	**Parameter**	**Rating (out of 10)**
1	Project Novelty: Potential of the project to create new knowledge	7.60
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	7.90
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	7.80
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	8.70
5	Social Applicability: Project work intends to solve a practical problem.	7.90
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	7.90

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PAC Member (HOD/Chairperson) Name: Dr. Amit Kumar Thakur	UID: 25010	Recommended (Y/N): Yes
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Final Topic Approved by PAC: **To study the effect of oxidizer on combustion of propellant in hybrid rocket engine using computational analysis**

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PAC CHAIRPERSON Name: 11108::Dr. Ankur Bahl **Approval Date:** 08 Dec 2025

ABSTRACT

Hybrid rocket motors have attracted increasing attention in modern propulsion research because they combine selected advantages of solid and liquid rocket systems. In a hybrid rocket motor, the fuel is stored in solid form while the oxidizer is supplied separately in liquid or gaseous form. This arrangement improves operational safety, reduces accidental ignition risk, and allows thrust control through oxidizer flow regulation.

The present project investigates the computational performance of a hybrid rocket motor loaded with fuel additives. A 100N thrust class hybrid rocket motor was designed, analysed, and validated using paraffin wax as the primary fuel, aluminium as metallic additive, and Lithium Borohydride as nano additive. Liquid Oxygen (LOX) was selected as the oxidizer. A cylindrical fuel grain geometry and stainless-steel converging-diverging nozzle were adopted. The chamber was structurally designed for 25 bars.

The objectives of the project were to develop and validate a computational model for analysing hybrid rocket motor performance with various fuel-additive combinations and to quantify the effect of additives on fuel regression rate, combustion efficiency, and specific impulse. Computer Aided Design (CAD) was used for geometry development, while NASA CEA was used for thermochemical performance prediction. Comparative results from recent studies indicate that paraffin-based fuels with energetic additives can significantly improve propulsion output.

The developed motor achieved successful ignition and short-duration combustion of approximately 2 seconds. The estimated practical specific impulse was in the range of 230–240 seconds. The study demonstrates that computational methods combined with practical validation provide an effective pathway for future hybrid rocket development.

Keywords: Hybrid Rocket Motor, Paraffin Wax, Aluminium, LOX, Specific Impulse, CEA, Propulsion

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LIST OF SYMBOLS

Symbol	Meaning
F	Thrust
Isp	Specific Impulse
Pc	Chamber Pressure
At	Nozzle Throat Area
dt	Throat Diameter
$\dot{m}$	Mass Flow Rate
$\dot{r}$	Regression Rate
O/F	Oxidizer to Fuel Ratio
C*	Characteristic Velocity
Tc	Chamber Temperature

LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
CEA	Chemical Equilibrium with Applications
LOX	Liquid Oxygen
HTPB	Hydroxyl-Terminated Polybutadiene
$LiBH_4$	Lithium Borohydride
$LiBeH_4$	Lithium Beryllium Hydride
$LiAlH_4$	Lithium Aluminium Hydride
ABS	Acrylonitrile Butadiene Styrene
Al	Aluminium
N_2O	Nitrous Oxide
KNO_3	Potassium Nitrate
NH_4NO_3	Ammonium Nitrate

CHAPTER 1

INTRODUCTION

1.1 General Introduction

Rocket propulsion is one of the most important branches of aerospace engineering because it enables motion in atmospheric as well as space environments where conventional air-breathing engines cannot operate efficiently. Rockets generate thrust by ejecting high-velocity gases in the opposite direction according to Newton's third law of motion. Depending on propellant storage and combustion arrangement, rocket engines are generally classified as solid, liquid, and hybrid propulsion systems [2], [3], [4], [5], [25], [44]. Solid rocket motors are simple, compact, and capable of generating high thrust. However, once ignited, thrust control is difficult and shutdown is generally not possible. Liquid rocket engines provide better controllability, restart capability, and high efficiency, but they require complex tanks, feed systems, valves, and pumps. Hybrid rocket motors combine useful features of both systems by storing fuel in solid form while oxidizer is supplied separately [2], [3], [4], [5], [25], [44], [45], [46], [47], [48], [49], [50], [51]. Because the fuel and oxidizer remain separated until operation, hybrid rocket motors generally offer safer handling than premixed solid propellant systems. This characteristic has made them attractive for educational projects, sounding rockets, and reusable research vehicles [2]-[5], [25], [44].

1.2 Importance of Hybrid Rocket Motors

Hybrid rocket motors are increasingly important because they provide a balance between safety, operational flexibility, and practical performance. They can be throttled by regulating oxidizer flow, and combustion can be terminated by shutting off oxidizer supply. These advantages are difficult to achieve in conventional solid rocket motors [2], [3], [4], [5], [25], [44], [51]. Modern aerospace programs are investigating hybrid propulsion for suborbital launch systems, small satellite applications, green propulsion concepts, and academic test platforms. Several recent reviews confirm that hybrid propulsion remains a promising area of future development [2]-[5], [25].

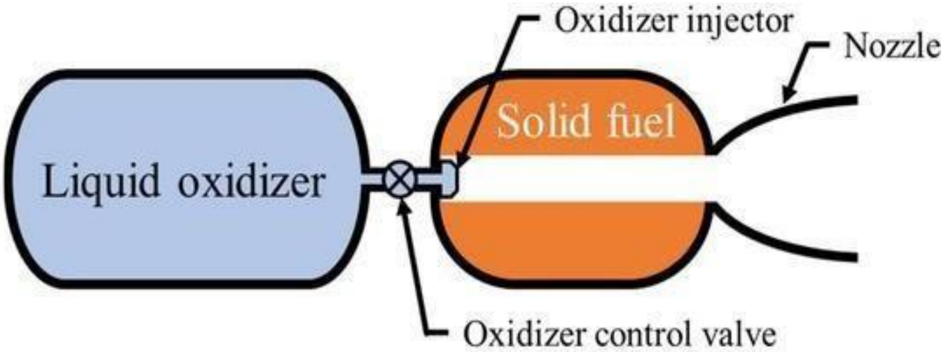


Figure 1. Basic schematic of a hybrid rocket propulsion system [25], [44]

1.3 Challenges in Hybrid Rocket Propulsion

Despite many advantages, hybrid rocket motors face technical challenges. The most common limitation is low fuel regression rate. Regression rate refers to the speed at which the fuel surface burns and recedes during operation. If the regression rate is low, fuel mass flow remains low and thrust generation is reduced [39]-[41], [47]-[49]. Other common challenges include incomplete combustion due to poor oxidizer-fuel mixing, chamber pressure fluctuation, nozzle erosion, thermal cracking of the fuel grain, and residue formation caused by metallic additives [45], [47]-[50] [47]. These limitations have encouraged researchers to investigate new fuels, additives, advanced grain geometry, optimized injectors, and computational modelling methods.

1.4 Need for Fuel Additives

Fuel additives are introduced into the base fuel to improve combustion behaviour and propulsion performance. Metallic additives such as aluminium release significant heat during combustion and may increase chamber temperature and theoretical specific impulse [1], [14], [27], [54]. Nano additives are of growing interest because their small particle size can increase reaction rate and combustion efficiency. Hydrogen-rich compounds such as Lithium Borohydride may improve exhaust velocity by reducing combustion product molecular weight. This makes them attractive for advanced propulsion studies [29], [41], [52].

1.5 Why Paraffin Wax Was Selected

Paraffin wax has become one of the most promising hybrid rocket fuels because it generally offers higher regression rate than many polymeric fuels such as Hydroxyl-

Terminated Polybutadiene. During combustion, paraffin forms a molten surface layer that can release droplets into the oxidizer stream, increasing burning intensity [11], [37], [49]. Paraffin is also inexpensive, easily available, simple to cast into grain shapes, and compatible with many additives. These practical advantages made it suitable for the present project [11], [32], [43].

1.6 Why Liquid Oxygen Was Selected

Liquid Oxygen (LOX) is a widely used rocket oxidizer because it provides strong oxidizing capability and supports high-energy combustion with hydrocarbon fuels. It is commonly used in professional launch systems due to its strong theoretical performance [2], [3], [4], [5], [25]. Although LOX handling requires cryogenic precautions, its performance advantages justified its use in the present project.

1.7 Role of Computational Tools

Modern propulsion development increasingly depends on computational tools. CAD software is used for component design and assembly verification. NASA CEA is widely used for thermochemical estimation of chamber temperature, specific impulse, and characteristic velocity. CFD tools help analyse internal flow behaviour, pressure distribution, and combustion zones [12], [39]-[41], [50], [58]. In the present project, CAD and CEA were directly used by the team, while CFD support was provided through collaboration.

1.8 Project Background

The present work involved the development of a 100 Newton thrust class hybrid rocket motor using paraffin wax fuel, aluminium additive, Lithium Borohydride nano additive, and Liquid Oxidizer. A cylindrical fuel grain geometry was selected because of manufacturing simplicity and predictable combustion behaviour. A stainless-steel converging-diverging nozzle was selected for gas expansion. The chamber was structurally designed for 25 bars. The project also aimed to demonstrate successful ignition and short-duration combustion.

1.9 Objectives of the Project

The main objective of the project was to develop and validate a computational model for analysing hybrid rocket motor performance with different fuel-additive combinations. Another objective was to quantitatively assess the influence of additives on fuel regression rate, combustion efficiency, and specific impulse. The project also aimed to design a compact 100 N class hybrid rocket motor, compare practical performance with theoretical predictions, and demonstrate successful ignition.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

Hybrid rocket propulsion has been widely studied in recent years due to increasing demand for safer, lower-cost, and efficient propulsion systems. Current research focuses on improving regression rate, additive loading, nozzle efficiency, injector design, and computational modelling. The references selected for this project provided the technical foundation for the present study [2]-[5], [9]-[12], [14]-[16], [24], [25], [39], [40], [41], [44]-[50].

2.2 Recent Development of Hybrid Rocket Propulsion

Hybrid rocket propulsion has experienced renewed global interest because it offers a practical compromise between solid and liquid propulsion systems. Recent review studies have shown that modern material technology, improved injector systems, additive manufacturing, and numerical modelling have significantly improved the feasibility of hybrid rocket motors for research and launch applications [2]-[5], [9], [24], [25]. Universities, private launch companies, and defense research groups are now using hybrid systems for educational demonstrations, sounding rockets, and prototype launch vehicles. Modern hybrid motors are especially attractive because they reduce handling risks associated with premixed solid propellants while avoiding some of the mechanical complexity of liquid rocket engines. These factors have contributed to rapid growth in hybrid propulsion research during the last decade [2]-[5], [25], [44]. Hydroxyl-Terminated Polybutadiene (HTPB) has historically been one of the most common fuels used in hybrid rocket motors. Researchers selected HTPB because it offers good structural strength, long storage life, and stable manufacturing characteristics. Studies involving HTPB with gaseous oxygen and nitrous oxide systems showed reliable combustion behaviour and acceptable thermal performance [33], [53]. However, multiple review papers concluded that HTPB often exhibits lower fuel regression rate than paraffin-based fuels under similar oxidizer mass flux conditions [44], [49]. Because of this limitation, researchers have attempted to improve HTPB performance by blending energetic additives or modifying grain geometry.

2.3 Studies on Paraffin Wax Fuel Systems

Paraffin wax has become one of the most promising fuels for hybrid propulsion because it generally provides higher regression rate than polymeric fuels. Experimental investigations on paraffin and paraffin-polyethylene wax blends showed improved thermal behaviour and favourable burning characteristics [11]. Researchers reported that paraffin forms a molten surface layer during combustion, and entrained droplets from this layer enhance burning intensity. This mechanism can significantly increase regression rate when compared with HTPB systems [37], [49]. Further studies on paraffin-based fuel blends such as paraffin-ethanol and paraffin-stearic acid systems also showed encouraging performance for laboratory hybrid motors [32], [43]. These findings strongly supported the selection of paraffin wax for the present project.

2.4 Studies on Metallic Additives

Metallic additives are widely used to improve energetic output in propulsion systems. Aluminium remains the most common additive because of its high heat of combustion. Computational analysis of metal additives in polymer fuels showed that aluminium can improve chamber temperature and theoretical specific impulse [1], [14], [27], [54]. Recent studies involving zirconium nanoparticles in boron-paraffin systems also demonstrated performance enhancement in ducted rocket and hybrid propulsion concepts [17]. Research on high entropy alloy additives has further indicated that advanced metallic additives may improve regression rate and combustion characteristics [4], [17], [41]. These studies confirm that metallic additives remain an important route for improving hybrid rocket performance. Nano additives have gained importance because their small particle size provides larger reactive surface area and faster combustion kinetics. Researchers studying carbon nanoparticles and energetic nano materials reported improvement in specific impulse and combustion efficiency [29], [41], [52]. Nano-scale additives are especially attractive in hybrid systems because they may improve ignition response, flame stability, and heat transfer to the fuel surface. This trend supports the investigation of Lithium Borohydride as a nano additive in the present project. Fuel grain geometry strongly affects burning area, fuel regression rate, chamber pressure behaviour, and thrust stability. Several recent studies showed that stepped helix ports, topology-optimized grains, and modified internal passages can significantly enhance hybrid rocket performance [9], [28], [35]. Experimental work on grain geometry modification also reported measurable thrust improvement in bench-top hybrid rocket engines [6]. However, although advanced grain shapes improve burning area, they may increase manufacturing complexity. For student-scale projects, cylindrical grains remain attractive because they are easier to cast, align, and analyse. Therefore, a cylindrical grain geometry was selected for the present work. Injector design directly influences oxidizer-fuel mixing and combustion efficiency. Poor injection can create local hot spots or incomplete combustion. Several recent studies reported that swirl injection can improve mixing, increase regression rate, and enhance combustion stability [16], [22], [30], [55]. Distributed tube injectors and optimized injector placement have also been studied for regression-rate enhancement [46]. These results demonstrate that injector design is one of the most critical aspects of hybrid rocket performance. The present project used a multi-hole injector arrangement to promote uniform oxidizer distribution. Nozzle geometry determines how efficiently chamber thermal energy is converted into thrust. Recent studies on nozzle expansion ratio optimization showed that proper nozzle sizing can significantly improve hybrid rocket efficiency [15], [24]. Numerical optimization of hybrid rocket nozzle geometry using finite volume methods also confirmed the importance of throat and exit dimensions [15]. Other studies involving nozzle erosion and thrust stand measurements demonstrated that nozzle material and chamber pressure strongly influence long-duration performance [45]. For short-duration educational systems, stainless steel remains a practical nozzle material.

2.5 CFD and Numerical Modelling Studies

Computational Fluid Dynamics has become essential in propulsion development. Recent studies simulated reacting flow inside N_2O/ABS hybrid rockets and predicted combustion behaviour successfully [12]. Other investigations focused on regression-rate modelling and flow-field prediction in hybrid rocket motors [39], [40], [41], [50], [58]. CFD is highly valuable because it reduces experimental cost and helps optimize injector geometry, chamber dimensions, and nozzle design before fabrication. In the present project, CFD support was performed through collaboration.

2.6 NASA CEA and Thermochemical Studies

Thermochemical analysis tools such as NASA CEA are widely used for early-stage rocket design. Recent studies used computational methods to screen fuel additives and compare energetic performance of paraffin systems enhanced by KNO_3, NH_4NO_3, aluminium, titanium, and other compounds [1], [14]. CEA allows rapid estimation of chamber temperature, exhaust species, characteristic velocity, and specific impulse. This makes it highly useful for comparing multiple additive combinations before practical testing. Recent experimental studies have reported successful testing of 200N class and 300N class hybrid rocket engines [3], [23]. These studies demonstrated that small thrust-class motors are valuable for validating combustion behaviour, injector performance, and hardware reliability. The present 100N class hybrid rocket motor belongs to the same practical development category and is highly suitable for academic project work.

2.7 Research Gaps Identified

The reviewed literature revealed several gaps that motivated the present project. First, limited open literature exists on Lithium Borohydride-loaded paraffin systems in small hybrid motors. Second, many studies focus only on experiments or only on numerical work, while combined practical and computational validation is less common. Third, low-cost student-scale motors still need more documented data on additive influence, short burn validation, and nozzle optimization.

2.8 Summary of Literature Review

The literature clearly shows that paraffin wax is one of the strongest candidate fuels for hybrid rocket systems because of its favourable regression rate. Metallic and nano additives can further improve performance. Grain geometry, injector design, nozzle sizing, and computational analysis are all critical to achieving efficient operation. The reviewed studies strongly supported the engineering decisions adopted in the present project.

CHAPTER 3

METHODOLOGY AND IMPLEMENTATION

3.1 Introduction

This chapter explains the methodology and practical implementation used for the development of the hybrid rocket motor. The project involved requirement definition, fuel and oxidizer selection, CAD modelling, combustion chamber design, nozzle sizing, thermochemical analysis, fabrication planning, and validation testing [2]-[5], [25], [44]-[50]. The project was carried out by a four-member team using a structured engineering design approach. At the initial stage, technical literature related to hybrid rocket propulsion, paraffin fuels, additives, nozzle design, and small thrust-class engines was studied in detail. This step was necessary to understand the current state of research and identify suitable design parameters for a student-scale hybrid rocket motor [25], [44].

After the literature review, the team finalized the target specifications of the system. The project was planned as a 100 N thrust class hybrid rocket motor with short duration burn capability, safe chamber pressure margin, and compatibility with available fabrication resources. A modular configuration was selected so that the chamber, grain section, and nozzle could be assembled and disassembled easily for inspection and maintenance. The overall design philosophy focused on simplicity, safety, computational validation, and practical feasibility.

3.2 Engineering Methodology

A sequential methodology was followed to complete the project successfully. The first step involved defining performance requirements such as thrust level, burn time, chamber pressure, and propellant combination. The second step involved selecting suitable fuel and oxidizer materials based on literature and availability.

The third step consisted of CAD modelling of the complete rocket motor assembly. Once geometry was finalized, chamber dimensions and nozzle parameters were calculated. NASA CEA simulations were then carried out to estimate theoretical specific impulse, combustion temperature, and optimum oxidizer-to-fuel ratio. After computational validation, hardware preparation and assembly planning were completed. The final step was short-duration ignition and validation testing.

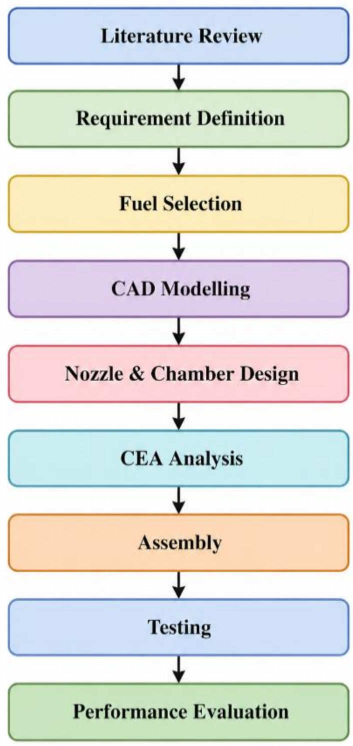


Figure 2.- Analytical framework for Hybrid Propulsion System Development

3.3 Selection of Fuel and Oxidizer

Paraffin wax was selected as the primary fuel because recent studies showed that paraffin generally provides higher regression rate than many polymeric fuels [11], [49]. Paraffin is also low-cost, easily available, and can be cast into cylindrical grains with comparatively simple processing.

Liquid Oxygen (LOX) was selected as the oxidizer because it provides strong oxidizing capability and high theoretical performance with hydrocarbon fuels [25]. Although LOX requires cryogenic handling procedures, its energetic benefits made it suitable for the present project. The chosen combination of paraffin wax and LOX represented a high-potential hybrid propellant pair for a compact experimental motor.

3.4 Selection of Additives

Aluminium was selected as the metallic additive because of its high heat release during combustion. Many published studies have shown that aluminium can increase combustion temperature and improve theoretical propulsion performance [1], [14], [27], [54].

Lithium Borohydride was selected as the nano additive because hydrogen-rich energetic compounds can improve specific impulse by generating lighter exhaust species. Lithium Borohydride also added research novelty to the project because fewer student-scale studies are publicly available on this material. The project therefore compared conventional metallic enhancement with advanced hydride-based enhancement.

3.5 Target Design Specifications

The final target specifications selected for the motor are given below.

Table 1. Final design specifications of the developed hybrid rocket motor

Parameter	Value
Design Thrust	100 N
Burn Duration	2 s
Chamber Design Pressure	25 bars
Oxidizer	LOX
Fuel	Paraffin Wax
Additives	Aluminium, Lithium Borohydride
Nozzle Type	Converging-Diverging
Nozzle Material	Stainless Steel
Grain Geometry	Cylindrical

These specifications were selected to balance safety, manufacturability, and meaningful propulsion output.

3.6 CAD Modelling of the Rocket Motor

CAD modelling was used to convert conceptual design into a manufacturable hardware layout. The complete motor assembly included oxidizer inlet section, combustion chamber body, fuel grain cavity, flange joints, and converging-diverging nozzle. CAD modelling helped verify alignment between parts, fitment of the fuel grain, sealing interfaces, and nozzle placement. It also supported dimensional checks before fabrication and served as the basis for future CFD work.

1. LOX Inlet

2. Chamber Body

3. Fuel Grain Section

4. Flange Connection

5. CD Nozzle

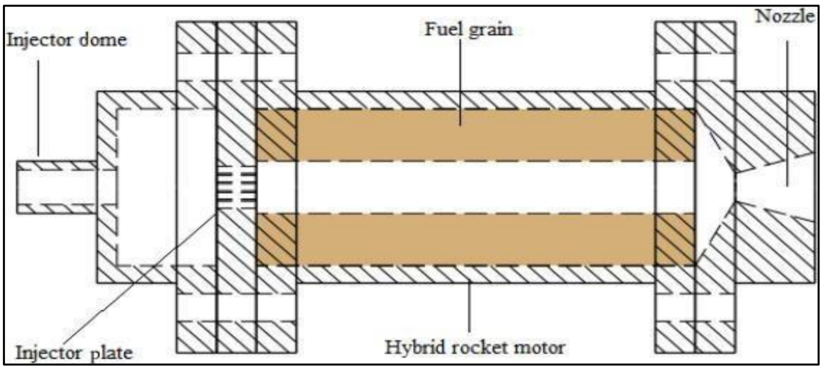


Figure 3. Cross-sectional mechanical drawing of the hybrid rocket engine [3], [25]

3.7 Finalized Geometrical Dimensions

The important geometrical dimensions extracted from the finalized design are given below.

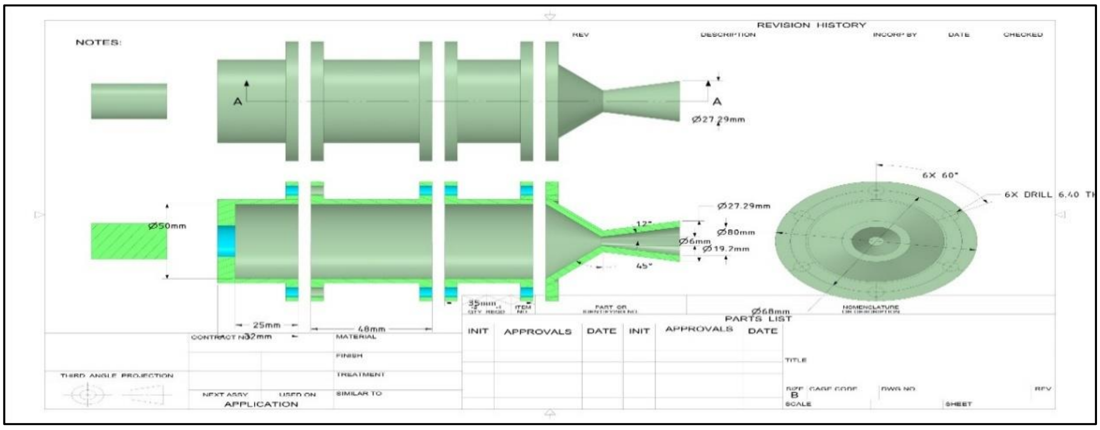


Figure 4. Complete assembly blueprint of the hybrid rocket motor

Table 2. Finalized geometrical dimensions of the developed hybrid rocket motor

Component	**Dimension**
Chamber Outer Diameter	68 mm
Flange Outer Diameter	80 mm
Chamber Length	48 mm
Front Section Length	25 mm
Fuel Grain Outer Diameter	24 mm
Fuel Port Diameter	12 mm
Injector Holes	6 × 2 mm
Nozzle Throat Diameter	6 mm
Nozzle Exit Diameter	27.47 mm
Throat Area	28.27 mm^2

These dimensions were selected to maintain compactness while supporting the required pressure and thrust class.

3.8 Combustion Chamber Design

The combustion chamber was one of the most critical components because it had to contain high-pressure combustion gases safely. For this reason, the chamber was structurally designed for 25 bars. Designing above working pressure provides safety margin and reduces structural risk.

Stainless steel was selected for the chamber because it offers good strength, corrosion resistance, and availability. Since the burn duration was short, stainless steel was suitable for thermal loading in the present application.

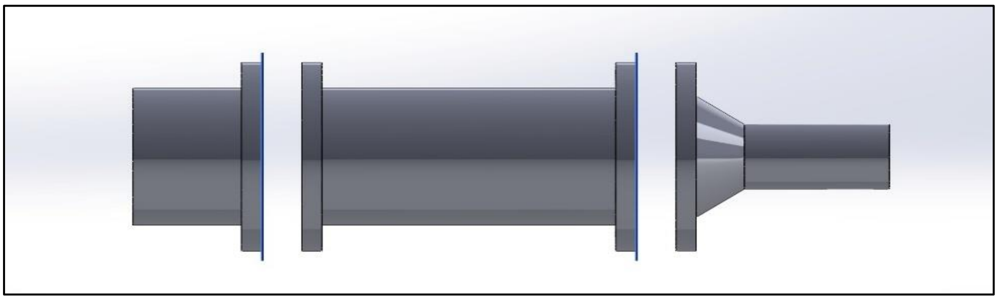


Figure 5. Isometric CAD representation of the hybrid rocket motor assembly

3.9 Fuel Grain Design

The fuel grain was cylindrical with an outer diameter of 24 mm and a central port diameter of 12 mm. A cylindrical grain was selected because it is easier to cast and machine than complex star-port or helical designs. It also provides relatively stable and predictable combustion behaviour.

During combustion, the inner port surface burns progressively outward. As the port diameter increases, the internal flow area changes, which influences regression rate and oxidizer mass flux.

3.10 Injector Design

The injector arrangement consisted of six holes of 2 mm diameter positioned to distribute oxidizer across the chamber inlet. Multi-hole injectors are commonly used in compact motors because they improve oxidizer distribution and mixing quality [16], [48]. Better oxidizer distribution supports stable ignition and more uniform combustion over the fuel port surface.

3.11 Nozzle Design

A converging-diverging nozzle was selected because it is the standard configuration for rocket propulsion. Combustion gases accelerate in the converging section, reach sonic speed at the throat, and then expand to higher velocity in the diverging section.

The throat diameter was fixed at 6 mm.

$$A_t = \frac{\pi d_t^2}{4}$$

For a throat diameter of 6 mm, the throat area was calculated as 28.27 mm². The exit diameter of 27.47 mm provided sufficient expansion area for the targeted operating condition.

Recent nozzle optimization studies support the importance of proper throat and expansion ratio selection in hybrid motors [15], [24].

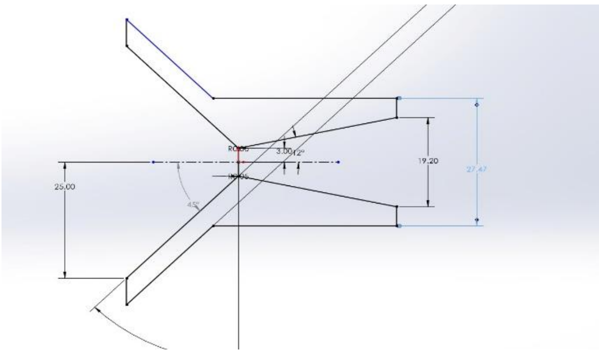


Figure 6. Dimensioned nozzle configuration used in the hybrid rocket motor

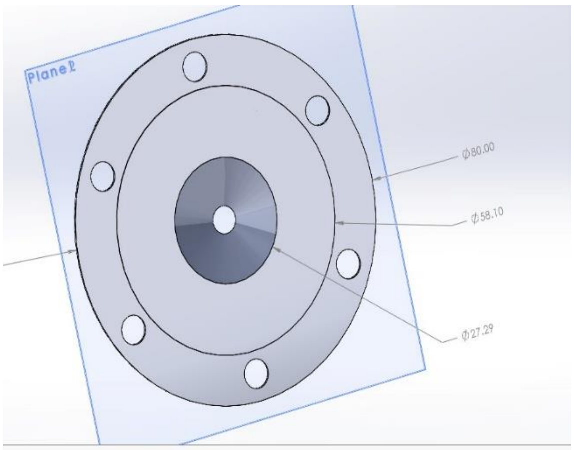


Figure 7. Dimensioned injector plate geometry used in the hybrid rocket motor

3.12 Design Calculations and Performance Methodology

3.12.1 Thrust and Mass Flow Rate Calculation

To achieve the design thrust of 100 N, the required total mass flow rate was computed using:

$$\dot{m} = \frac{T}{I_{sp}\, g_0}$$

3.12.2 Oxidizer Mass Flux (G_{ox})

Oxidizer mass flux governs the rate of heat transfer into the fuel surface and directly influences the regression rate of paraffin-based fuels. Higher values of G_{ox} strengthen convective heating and promote stronger melt-layer entrainment, especially for aluminium-enhanced formulations [39]. It is one of the most critical parameters determining overall fuel consumption in hybrid systems [40].

Equation:

$$GOX = \frac{\dot{m}_0 x}{Aport}$$

Here, (Gox) represents the oxidizer mass flux and is expressed in $kg \cdot m^{-2} \cdot s^{-1}$. The term ($\dot{m}\, ox$) denotes the oxidizer mass flow rate, measured in $kg \cdot s^{-1}$, while ($Aport$) refers to the fuel port cross-sectional area, expressed in m^2.

3.12.3 Regression Rate ($\dot{r}$)

Regression rate defines how quickly the solid fuel surface recedes during combustion. Paraffin exhibits a strong dependence on oxidizer mass flux, resulting in higher regression rates than traditional polymeric fuels [41]. Nano-aluminium additives intensify heat feedback, increasing the value of $\dot{r}$ even further [42].

Equation:

$$\dot{r} = a\, G_{ox}^{n}$$

Here, ($\dot{r}$) represents the fuel surface regression rate or surface recession rate, measured in $m \cdot s^{-1}$. The terms (a) and (n) are empirical constants obtained from experimental studies and reported in the literature [43]. The parameter (Gox) denotes the oxidizer mass flux, expressed in $kg \cdot m^{-2} \cdot s^{-1}$.

3.12.4 Thrust (T)

Thrust represents the net force produced by the hybrid rocket due to exhaust momentum and pressure difference across the nozzle. This parameter determines the mass flow required to meet the targeted 100 N and is essential for evaluating how fuel formulation affects performance [44].

Equation:

$$\boldsymbol{T = \dot{m}_{ve} + (p_e - p_a)Ae}$$

Here, (T) represents the thrust produced by the rocket engine and is measured in newtons (N). The term ($\dot{m}$) denotes the total mass flow rate, expressed in $kg \cdot s^{-1}$, while (ve) represents the exhaust velocity in $m \cdot s^{-1}$. The parameters (Pe) and (Pa) indicate the nozzle exit pressure and ambient pressure, respectively, both measured in pascals (Pa). The term (Ae) refers to the nozzle exit area, expressed in m^2.

3.12.5 Specific Impulse (I_{sp})

Specific impulse measures the efficiency of the propellant combination. For LOX–paraffin systems, values around 230 s are typical, while aluminium additions can slightly enhance gas-phase energy release [45].

Equation:

$$\boldsymbol{I_{sp} = \frac{T}{\dot{m}g_0}}$$

Here, (I_{sp}) represents the specific impulse of the propulsion system and is measured in seconds (s). The parameter (T) denotes the thrust produced by the engine, expressed in newtons (N). The term ($\dot{m}$) refers to the mass flow rate, measured in $kg \cdot s^{-1}$, while (g_0) represents the standard gravitational acceleration, equal to 9.80665 $m \cdot s^{-2}$.

3.12.6 Effective Exhaust Velocity (v_e)

The exhaust velocity is determined by chamber temperature, gas properties, and nozzle expansion. Aluminium nanoparticles raise combustion temperature, which typically increases exhaust velocity [46].

Equation:

$$v_e = \sqrt{\frac{2\gamma}{\gamma - 1} R T_c \left(1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma-1}{\gamma}}\right)}$$

Here, (v_e) represents the exhaust velocity and is measured in m·s⁻¹. The symbol (γ) denotes the specific heat ratio of the combustion gases. The parameter (R) represents the specific gas constant, expressed in $J \cdot kg^{-1} \cdot K^{-1}$, while ($T_c$) denotes the combustion chamber temperature in kelvin (K). The terms (P_c) and (P_e) represent the chamber pressure and nozzle exit pressure, respectively, both measured in pascals (Pa).

3.12.7 Characteristic Velocity (c*)

Characteristic velocity indicates combustion quality and is independent of nozzle performance. Higher values reflect efficient chemical energy release [47].

Equation:

$$C^* = \frac{P_C A_t}{m}$$

Here, (c*) represents the characteristic velocity and is measured in $m \cdot s^{-1}$. The parameter (Pc) denotes the chamber pressure, expressed in pascals (Pa). The term (At) refers to the nozzle throat area, measured in m^2, while (m) represents the total mass flow rate in $kg \cdot s^{-1}$.

3.12.8 Throat Area (A_t)

Correct throat sizing ensures proper mass flow without causing excessive chamber pressure. This is one of the primary geometric parameters for nozzle design [48].

Equation:

$$A_t = \frac{\dot{m} C^*}{P_C}$$

Here, $A_t$represents the nozzle throat area and is measured in m^2. The term $\dot{m}$ denotes the total mass flow rate, expressed in $kg \cdot s^{-1}$. The parameter c^*refers to the characteristic velocity, measured in $m \cdot s^{-1}$, while $P_c$represents the chamber pressure in pascals (Pa).

3.12.9 Throat Diameter (d_t)

The throat diameter directly affects manufacturability and erosion behaviour, especially when metallic additives are present in the exhaust [49].

Equation:

$$\boldsymbol{d_t = \sqrt{\{\{4A_t\}/\{\pi\}\}}}$$

Here, (d_t) represents the nozzle throat diameter and is measured in metres (m). The parameter (A_t) denotes the nozzle throat cross-sectional area, expressed in square metres (m^2).

3.12.10 Chamber Pressure (Pc)

Chamber pressure is determined from mass flow and throat geometry. Higher aluminium content may increase Pc due to improved burning rates [50].

Equation:

$$\boldsymbol{P_C = \frac{\dot{m}C^*}{A_t}}$$

Here, (P_c) represents the chamber pressure and is measured in pascals (Pa). The term ($\dot{\boldsymbol{m}}$) denotes the total mass flow rate, expressed in $kg \cdot s^{-1}$. The parameter (c^*) refers to the characteristic velocity, measured in $m \cdot s^{-1}$, while (A_t) represents the nozzle throat cross-sectional area in m^2.

3.12.11 Combustion Efficiency (ηc)

Combustion efficiency measures how close real performance is to theoretical CEA estimates. Hybrid rockets often show moderate losses due to limited mixing, though metal additives can improve performance [52].

Equation:

$$\boldsymbol{n_c = \left(\frac{c_{real^*}}{cth}\right) \times 100}$$

where

$$\boldsymbol{c_{real^*} = \frac{p_C A_t}{m}}$$

Here, (ηc) represents the combustion efficiency and is usually expressed as a percentage (%). The terms (creal*) and (ctheoretical *) denote the real (experimental) and theoretical characteristic velocities, respectively, both measured in $m \cdot s^{-1}$.

3.12.12 Total Mass Flow Rate ($\dot{m}$)

Mass flow determines how much propellant is required to reach 100 N using the assumed specific impulse of 230 s.

Equation:

$$\dot{\boldsymbol{m}} = \frac{\boldsymbol{T}}{\boldsymbol{Isp g_0}}$$

Here, ($\dot{\boldsymbol{m}}$) represents the total mass flow rate and is measured in $kg \cdot s^{-1}$. The parameter (T) denotes the thrust produced by the engine, expressed in newtons (N). The term ($\boldsymbol{Isp}$) refers to the specific impulse, measured in seconds (s), while (g_o) represents the standard gravitational acceleration in $m \cdot s^{-2}$.

3.12.13 Fuel & Oxidizer Flow

The oxidizer–fuel distribution is determined by the selected O/F ratio. Paraffin–LOX typically performs well between 4 and 6.

Equation:

$$\dot{\boldsymbol{m}}_f = \frac{\dot{\boldsymbol{m}}}{1 + \frac{\boldsymbol{O}}{\boldsymbol{F}}}$$

$$\dot{\boldsymbol{m}}_{ox} = \frac{(\boldsymbol{O/F})\,\dot{\boldsymbol{m}}}{1 + \boldsymbol{O/F}}$$

Here, ($\dot{m}_{\boldsymbol{f}}$) represents the fuel mass flow rate and is measured in $kg \cdot s^{-1}$. The term ($\dot{\boldsymbol{m}}_{\boldsymbol{ox}}$) denotes the oxidizer mass flow rate, also expressed in $kg \cdot s^{-1}$. The parameter (O/F) refers to the oxidizer-to-fuel ratio, which indicates the proportion of oxidizer mass flow to fuel mass flow during combustion.

Table 3. Design and performance parameters of the developed hybrid rocket motor

Parameter	**Symbol**	**Value**	**Units**
Design Thrust	T	100	N

Total Mass Flow Rate	$\dot{m}$	0.0443	kg/s
Fuel Flow Rate	$\dot{m}f$	0.00738	kg/s
Oxidizer Flow Rate	$\dot{m}ox$	0.03692	kg/s
Fuel Port Area	Aport	1.131×10^{-4}	m^2
Oxidizer Mass Flux	Gox	326.4	$kg/m^2 \cdot s$
Regression Rate	$\dot{r}$	4.52	mm/s
Throat Diameter	dt	6	mm
Throat Area	At	28.27	mm^2
Operating Chamber Pressure	Pc	20.4	bar
Structural Design Pressure	Pc,max	25	bar
Specific Impulse	Isp	230	s
Exhaust Velocity	ve	2255.5	m/s
Combustion Temperature	Tc	2800	K

3.13 NASA CEA Analysis Procedure

NASA CEA was used as the primary thermochemical tool for comparing fuel combinations. Inputs included chamber pressure, oxidizer type, and fuel chemistry. The software provided outputs such as chamber temperature, characteristic velocity, and specific impulse. This method allowed rapid comparison between paraffin blended with aluminium, paraffin blended with Lithium Borohydride, and other energetic fuel cases without expensive repeated testing.

3.14 Fabrication and Assembly Considerations

After finalizing the design, components were prepared for assembly. Chamber body, nozzle, and flange interfaces were aligned carefully to reduce leakage risk. Proper sealing was important because pressure losses can reduce performance. Fuel grain fitment also required dimensional accuracy to ensure a centered internal port and stable oxidizer flow path.

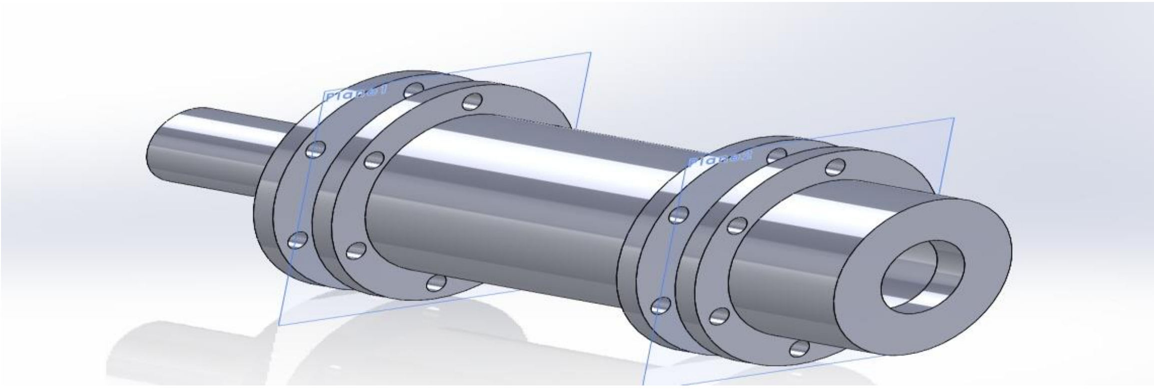


Figure 8. Isometric CAD model of the assembled hybrid rocket motor

3.15 Testing and Validation Procedure

The assembled motor was mounted securely for testing. Standard precautions such as controlled ignition distance, proper support fixtures, and safe supervision were followed. The motor achieved successful ignition and approximately 2 seconds of combustion.

This result validated the practical feasibility of the developed hybrid rocket system.

3.16 Challenges Faced During Implementation

The project involved practical challenges such as additive handling, precision fabrication, sealing reliability, ignition timing, and balancing theoretical design with available resources. Such constraints are common in student propulsion projects.

Despite these limitations, the project successfully progressed from concept stage to functional validation.

3.17 Summary of Chapter

This chapter described the complete methodology and implementation process used in the project. A paraffin-based additive-loaded hybrid rocket motor was designed using CAD tools, analyzed using NASA CEA, and validated through successful ignition testing. The adopted workflow reflected a systematic aerospace engineering approach.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter presents the results obtained from computational analysis and practical validation of the developed hybrid rocket motor. The discussion includes additive comparison, expected thermochemical performance, practical motor behaviour, and engineering interpretation of the findings.

4.2 Practical Validation Results

The most significant outcome of the project was the successful practical validation of the developed hybrid rocket motor. During testing, the motor achieved successful ignition and sustained combustion for approximately 2 seconds. This confirmed that the selected

Figure 9. Fabricated prototype model of the developed hybrid rocket motor

propellant combination, chamber design, injector arrangement, and nozzle geometry were suitable for short-duration operation. For a student-scale propulsion project, reliable ignition is an important milestone because ignition instability is one of the most common challenges in hybrid rocket systems. The observed result demonstrated that the system could initiate combustion and maintaining stable chamber conditions during the intended burn period. The estimated practical specific impulse was in the range of 230–240 seconds, which is technically reasonable for a compact hybrid rocket motor considering real-world performance losses.

4.3 Designed Thrust Performance

The motor was designed as a 100N thrust class hybrid rocket motor. This thrust level was selected because it is appropriate for academic testing, manageable hardware dimensions, and reduced operational risk compared with larger systems.

A 100 N motor is sufficiently powerful to demonstrate key propulsion phenomena such as chamber pressurization, nozzle acceleration, fuel regression, ignition behaviour, and additive influence. The achieved practical outcome validated the original design objective.

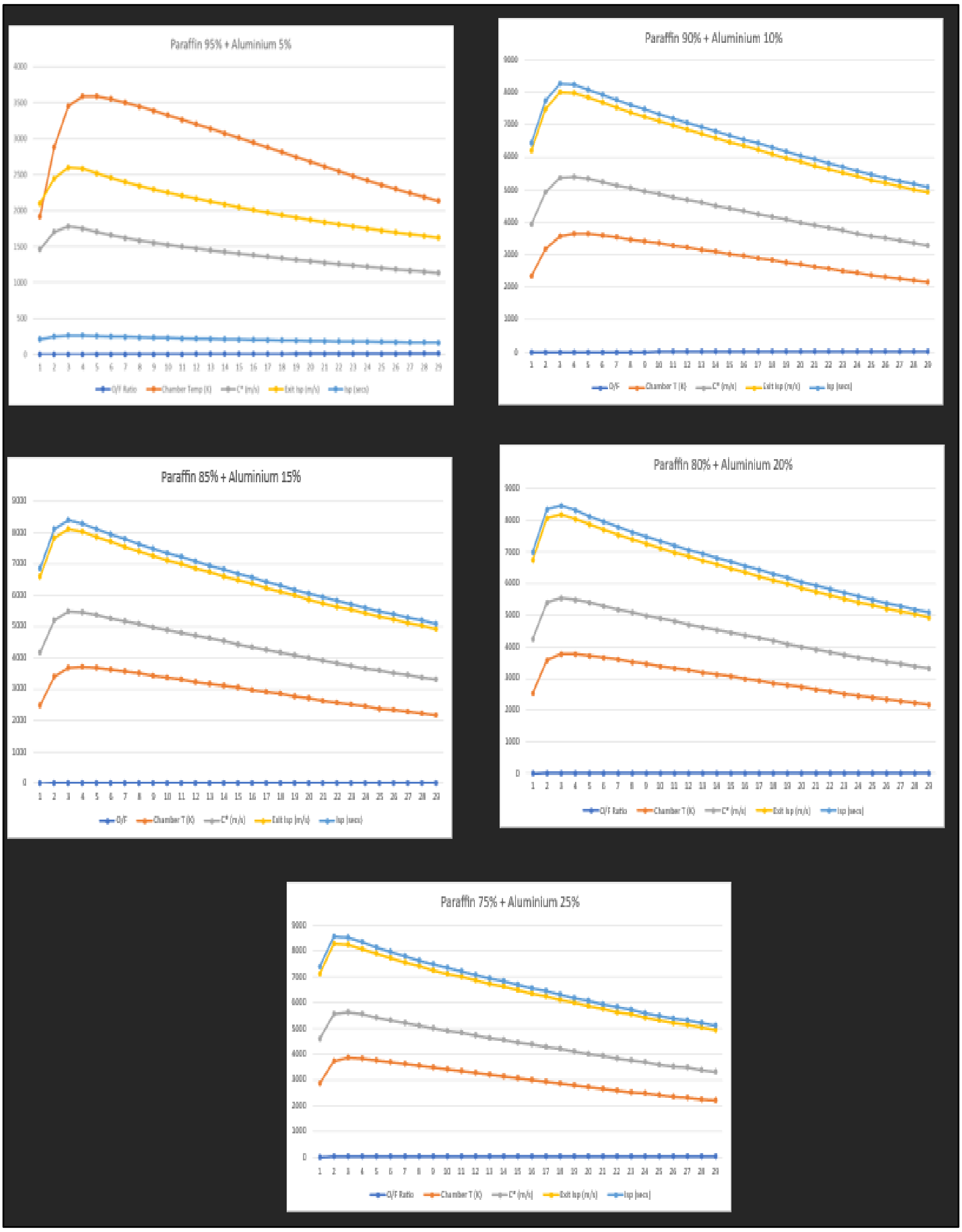
4.4 NASA CEA Comparative Performance Results

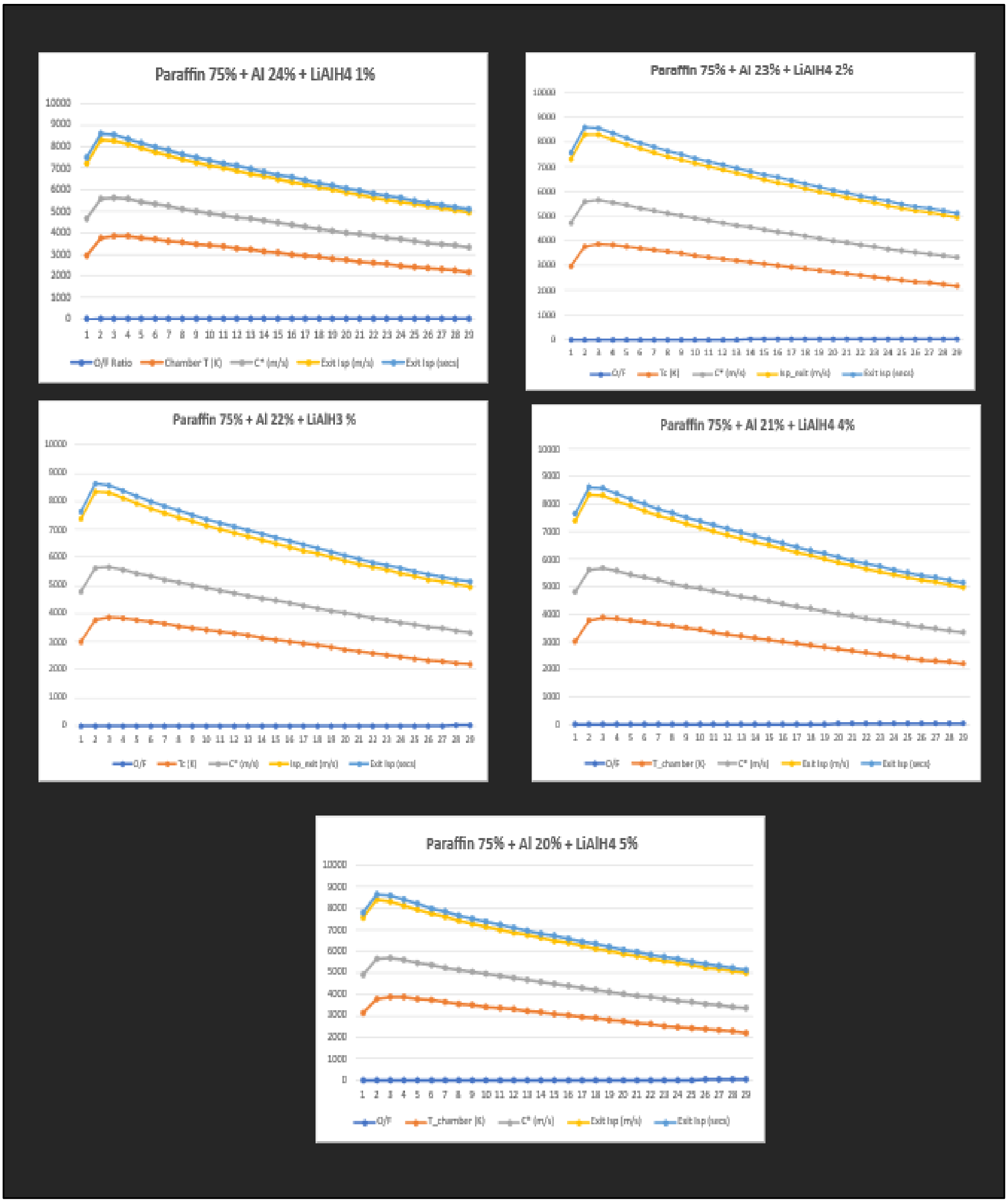
NASA CEA analysis was performed to compare different paraffin-based fuel combinations containing energetic additives. The highest predicted performance values from the study are summarized below.

Table 4. Comparative thermochemical performance of paraffin fuel blends with additives

Fuel Combination	Maximum Specific Impulse (s)	Maximum Chamber Temperature (K)	Maximum C* (m/s)
Paraffin + Aluminium	275.92	3844.78	1842.3
Paraffin + Aluminium	279.78	3871.60	1865.4
Paraffin + $LiAlH_4$	278.62	3879.91	1856.9
Paraffin + $LiBeH_4$	280.75	3891.83	1870.0

Figure 10. Performance analysis of Paraffin with Aluminium




Paraffin 75% + Al 24% + LiAlH4 1%
O/F Ratio
Chamber T (K)
C* (m/s)
Exit Isp (m/s)
Exit Isp (secs)
Paraffin 75% + Al 23% + LiAlH4 2%
O/F
Tc (K)
C* (m/s)
Isp_exit (m/s)
Exit Isp (secs)
Paraffin 75% + Al 22% + LiAlH3 %
O/F
Tc (K)
C* (m/s)
Isp_exit (m/s)
Exit Isp (secs)
Paraffin 75% + Al 21% + LiAlH4 4%
O/F
T_chamber (K)
C* (m/s)
Exit Isp (m/s)
Exit Isp (secs)
Paraffin 75% + Al 20% + LiAlH4 5%
O/F
T_chamber (K)
C* (m/s)
Exit Isp (m/s)
Exit Isp (secs)

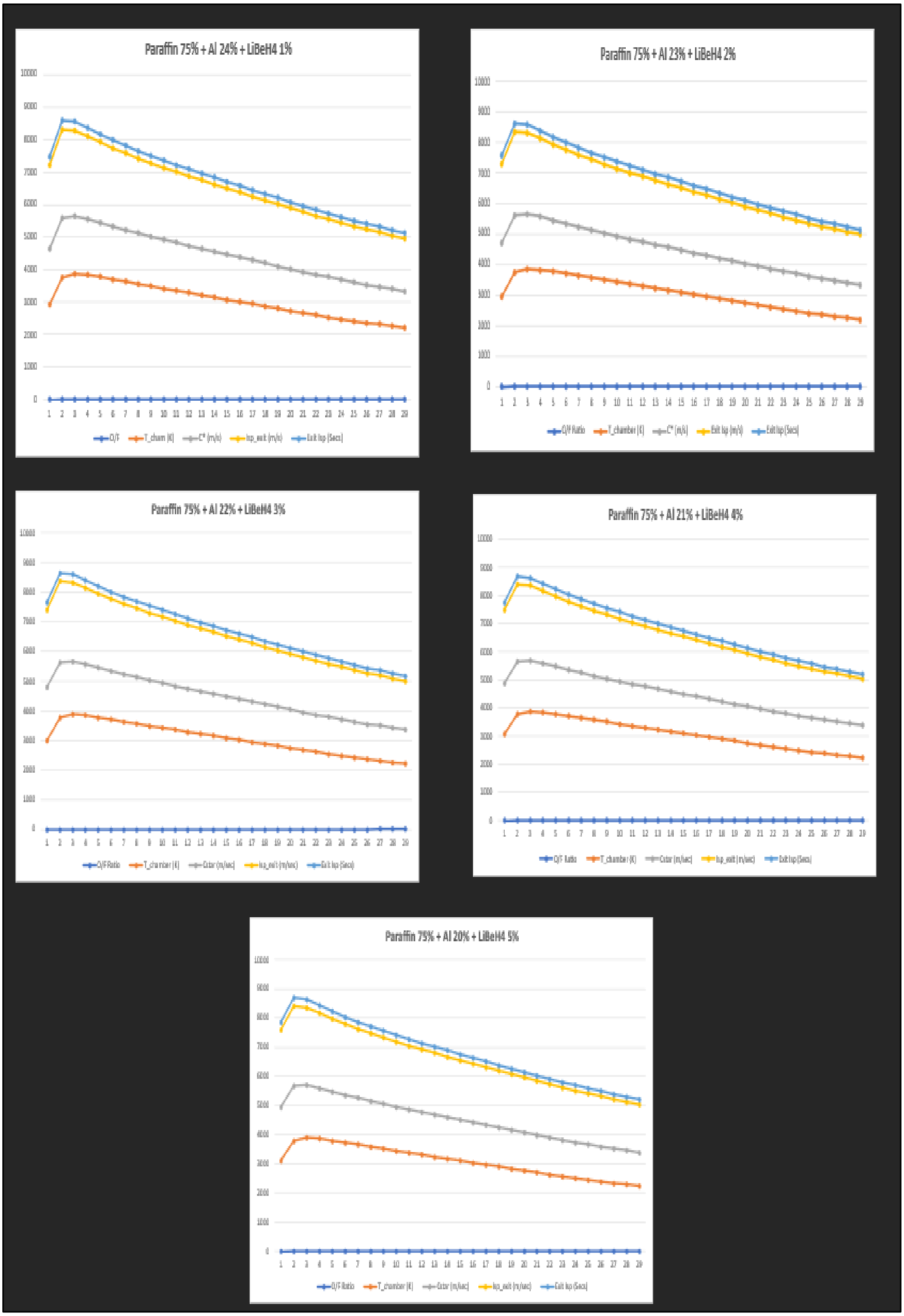


Figure 11. Performance analysis of Paraffin + Aluminium + LiBeH4

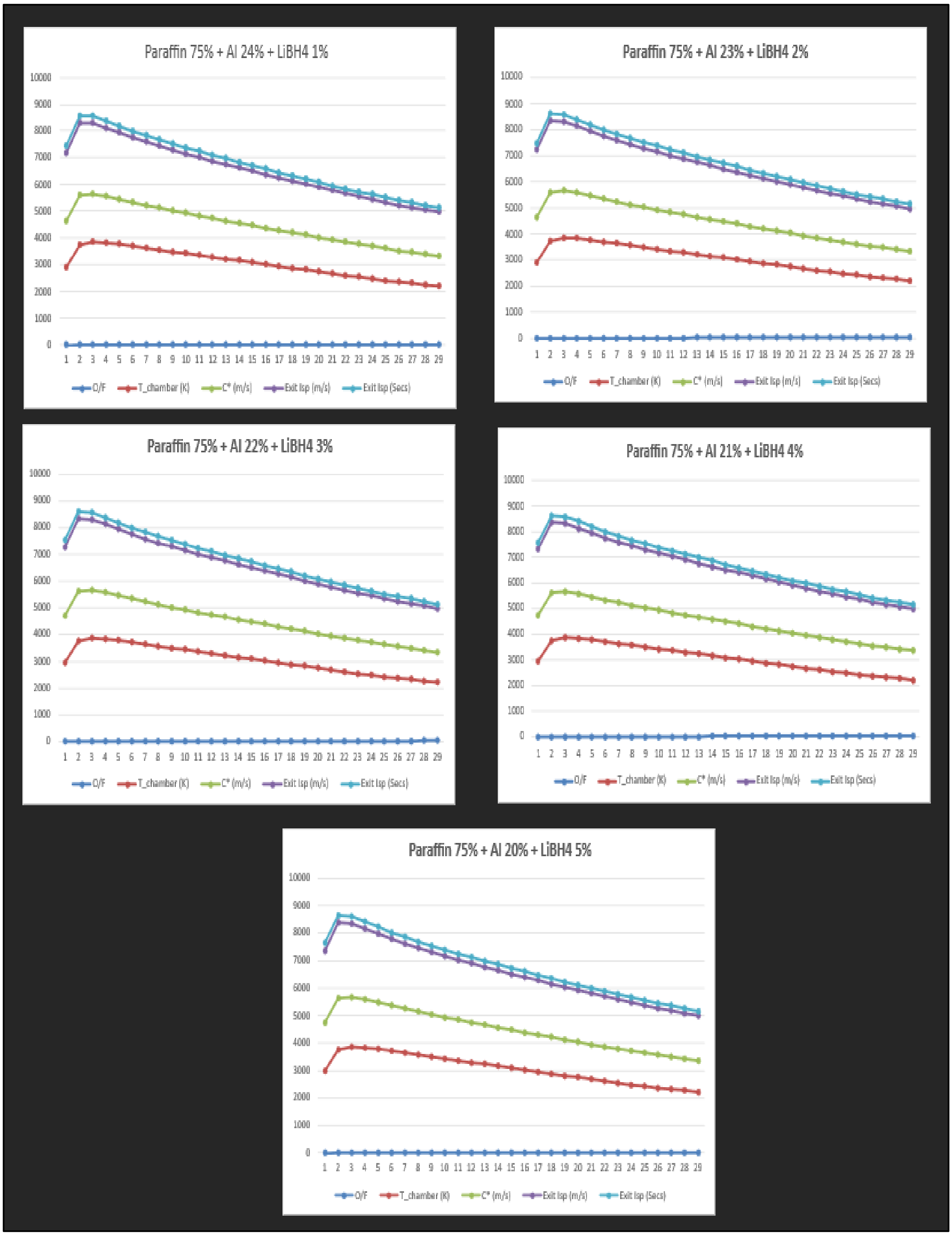


Figure 12. Performance analysis of Paraffin + Aluminium + LiBH4

The results clearly indicate that additive selection significantly influences theoretical hybrid rocket performance.

4.5 Interpretation of Additive Effects

The paraffin blended with aluminium combination showed strong performance because aluminium contributes high combustion energy and elevated flame temperature. This aligns with recent computational and experimental studies on energetic metallic additives [1], [27]. The paraffin blended with Lithium Borohydride combination produced a higher predicted specific impulse than paraffin blended with aluminium. This suggests that Lithium Borohydride provides benefits related to hydrogen-rich chemistry and lower exhaust molecular weight. Such behaviour is desirable because lower molecular weight combustion products generally increase exhaust velocity. Although paraffin blended with Lithium Borohydride showed the highest theoretical specific impulse among the compared cases, the present project focused on Lithium Borohydride because it offered strong performance while remaining suitable for the intended comparative study [29], [41], [52].

These findings suggest that hydride-based additives are promising candidates for future hybrid propulsion research.

4.6 Practical Performance vs Ideal Performance

The validated practical specific impulse of 230–240 seconds was lower than the ideal values near 275–280 seconds predicted by CEA. This difference is expected because ideal thermochemical software assumes complete combustion, equilibrium chemistry, no heat loss, and perfectly efficient nozzle expansion. Real systems experience several inefficiencies such as chamber wall heat transfer, injector pressure losses, incomplete oxidizer-fuel mixing, startup transients, nozzle losses, and short-duration instability. Therefore, the practical result obtained in the project remains realistic and technically acceptable. Many published small hybrid rocket studies report practical values below ideal predictions for the same reason [23], [26], [59].

4.7 Effect of Chamber Pressure

The chamber was structurally designed for 25 bars. This approach provided an important engineering safety margin. Designing pressure-containing hardware above expected operating conditions is common practice because it improves reliability and reduces failure probability. Chamber pressure directly affects combustion rate, gas density, and nozzle thrust generation. In general, higher chamber pressure can improve performance if the structure and feed system are capable of safely supporting it.

4.8 Effect of Nozzle Throat Diameter

The nozzle throat diameter was fixed at 6 mm. Throat diameter is one of the most critical design parameters because it governs choked mass flow rate and chamber pressure.

If the throat is too small, chamber pressure can rise excessively and structural loading increases. If it is too large, chamber pressure may fall and thrust output decreases. The selected 6 mm throat provided a practical balance for the targeted 100 N class system. The throat area of a converging-diverging nozzle plays a critical role in controlling chamber pressure and engine performance. A smaller throat area increases chamber pressure due to restricted mass flow, enhancing combustion but imposing higher thermal and structural loads. In contrast, a larger throat area reduces chamber pressure by allowing greater flow expansion, which may lower thrust efficiency. Therefore, an optimized throat size is essential to achieve stable thrust output while maintaining efficient and safe operating conditions.

Recent nozzle optimization studies also confirm the importance of correct throat sizing in hybrid motors.

4.9 Fuel Grain Behaviour

The cylindrical grain with 12 mm central port performed suitably for the intended short-duration application. Cylindrical grains are easier to manufacture and provide a stable internal flow passage. During combustion, the port diameter increases gradually as the fuel surface regresses outward. Although advanced grain geometries such as helical or stepped ports may improve burning area and regression rate, the cylindrical design offered better simplicity, lower manufacturing complexity, and reliable performance for the present project.

4.10 Burn Duration Analysis

The burn duration of approximately 2 seconds was sufficient for proof-of-concept validation. Short burn times are common in student projects because they reduce thermal loading on the chamber and nozzle while allowing ignition and combustion behaviour to be evaluated safely. Longer burn durations in future systems would require improved oxidizer feed control, increased fuel mass, thermal protection, and more advanced instrumentation.

4.11 Comparison with Published Studies

Recent small hybrid rocket studies involving 200 N and 300 N class engines demonstrated successful operation with practical limitations similar to those observed in student-scale systems [3], [23]. The present 100 N class motor achieving successful ignition and practical specific impulse of 230–240 seconds compares favourably with the performance range reported for compact hybrid rocket systems. This indicates that the developed project achieved technically credible results within its intended scale.

4.12 Strengths of the Present Project

The project demonstrated several engineering strengths. The use of paraffin wax supported favourable combustion behaviour. The use of aluminium and Lithium Borohydride introduced advanced additive-based performance enhancement concepts. The modular chamber assembly improved maintenance capability. The 25 bars structural design improved safety margin. CAD modelling and CEA analysis strengthened the scientific basis of the design process. These strengths make the project academically valuable and practically relevant.

4.13 Limitations of the Present Work

Although the project achieved its objectives, certain limitations remained. Full thrust-time instrumentation was limited. Long-duration hot-fire testing was outside the immediate scope. CFD was conducted through collaboration rather than fully integrated within the core team. Additive dispersion consistency may also influence repeatability. These limitations are common in university-scale propulsion projects and provide direction for future improvement.

4.14 Summary of Chapter

This chapter showed that the developed 100 N hybrid rocket motor produced meaningful technical results. Successful ignition, practical specific impulse of 230–240 seconds, and supportive thermochemical predictions confirmed the feasibility of paraffin-based additive-loaded hybrid propulsion. The results also indicated that Lithium Borohydride is a promising additive for future.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The present project successfully investigated the computational and practical performance of a hybrid rocket motor loaded with fuel additives. A 100 Newton thrust class hybrid rocket motor was designed using paraffin wax as the base fuel, aluminium as metallic additive, Lithium Borohydride as nano additive, and Liquid Oxygen as oxidizer.

The complete system was developed through a structured engineering process involving literature review, CAD modelling, chamber design, nozzle sizing, and thermochemical analysis using NASA CEA. The combustion chamber was designed for 25 bars. A stainless-steel converging-diverging nozzle with 6 mm throat diameter was used.

Comparative analysis showed that paraffin fuels containing advanced additives can provide higher ideal specific impulse than conventional systems. The paraffin blended with Lithium Borohydride combination demonstrated strong theoretical performance, supporting the additive selection. Practical testing resulted in successful ignition and approximately 2 seconds of combustion. The estimated practical specific impulse of 230–240 seconds confirmed that the developed motor was capable of useful propulsion output.

Therefore, the project successfully achieved its objectives of computational modelling, additive evaluation, and practical feasibility demonstration.

5.2 Major Findings

The study confirmed that paraffin wax is a suitable fuel for student-scale hybrid rocket systems because of favourable burning characteristics and ease of handling. Aluminium improved energetic output, while Lithium Borohydride showed strong thermochemical potential. Proper chamber pressure margin and nozzle throat sizing were essential for stable operation. Computational tools significantly reduced design uncertainty.

5.3 Future Scope

The present project opens several opportunities for future work. A calibrated thrust stand may be added to obtain direct thrust-time curves. Longer burn duration tests may be conducted using improved oxidizer feed systems and increased fuel loading.

Future systems may be scaled from 100 N to 250 N or 500 N thrust classes. Advanced injectors such as swirl injectors may improve combustion efficiency further. Graphite or refractory alloy nozzles may improve thermal durability. Future comparative studies may evaluate Lithium Borohydride against graphene, nano aluminium, boron compounds, or other advanced additives. Hybrid propulsion has promising future applications in sounding rockets, reusable research vehicles, educational launch systems, and economical access-to-space programs.

WORK PLAN TIMELINE

Sl.no.	Activity	Status
1	Topic selection and planning	Completed
2	Literature review	Completed
3	Fuel and additive selection	Completed
4	CAD modelling	Completed
5	Chamber and nozzle calculations	Completed
6	NASA CEA analysis	Completed
7	Assembly and preparation	Completed

8	Testing and ignition validation	Completed
9	Results analysis and report writing	Completed

CHAPTER 6

REFERENCES

[1] Ndlovu, G., Oboirien, B., & Ndungu, P. (2025). Screening of metal additives in ABS polymer fuel for enhanced performance in hybrid rocket motors: A computational analysis using CEA. Defence Technology.

[2] Roga, S. (2025). Modern rocket propulsion: a critical review of technological advances and ongoing challenges. International Journal of Chemical Reactor Engineering.

[3] Fabiani, M., Cassese, S., Capone, V. M., Migliorino, M. T., Mungiguerra, S., Bianchi, D., ... & Savino,R. (2025). Numerical and experimental analysis of a 200 N class GOX-ABS hybrid rocket engine. Aerospace Science and Technology, 111157.

[4] Sujana, M. J., Ismail, I. I., Nordin, N. H., & Azami, M. H. (2025). Investigation of Rate Enhancement of HTPB/Paraffin Fuel in the Hybrid Rocket Motor Utilizing High Entropy Alloys Energetic Additives. Journal of Advanced Research in Experimental Fluid Mechanics and Heat Transfer, 21(1), 23-37.

[5] Pawar, A., Praveen, A., Chavhan, H., & Thakur, A. K. (2025). Experimental and theoretical investigation of combustion efficiency, regression rate, and thrust characteristics of a hybrid rocket engine using PVC/DBP fuel. Aerospace Systems, 1-14.

[6] Kitchell, S., Abdelhalim, Y., Wehmeyer, A., Guy, S., & Keys, D. J. (2025). Improving Thrust Performance in Hybrid Rocket Engines through Fuel Grain Geometry Modification. In 2025 Regional Student Conferences (p. 100276).

[7] Carolan, M. (2025). Design, Development, and Testing of a Modular Bench-Top Hybrid Rocket Engine with Energetic Fuel (Master's thesis, South Dakota School of Mines and Technology).

[8] Khaskheli, M. A., Bhutto, A. A., Memon, K. N., Hussain, M., & Kalhoro, N. B. (2025). NUMERICAL ANALYSIS AND ENVIRONMENTAL IMPACT OF AFTERBURNING EXHAUST PLUMES IN ROCKET AND JET ENGINES. Spectrum of Engineering Sciences, 3(6), 929-950.

[9] Melis, M., Souza, A., & Afonso, F. (2025). On the multidisciplinary topology optimization of solid grain for a hybrid rocket. Engineering Optimization, 1-19.

[10] Charan, D. P., & Tibrewal, P. (2025, March). Modeling Thrust for ABS/N20 Based 3D Printed Hybrid Rocket Engine-Review of Static Fire Results. In 2025 IEEE Aerospace Conference (pp. 1-7). IEEE.

[11] Tian, H., Lu, Y., Li, L., Chen, R., Niu, X., Zhang, Y., & Cai, G. (2025). Experimental investigation on thermal behavior and regression rate characteristics of paraffin-polyethylene wax blends as hybrid rocket fuels. Acta Astronautica.

[12] Varathalingarajah, T., Harman, T., Birch, D., Garcia, B., & Novoselov, A. G. (2025). Reacting Flow Simulation of an N2O/ABS Hybrid Rocket. In AIAA SCITECH 2025 Forum (p. 2627).

[13] Boretti, A. (2025). Metal additive manufacturing for internal combustion engine components: a narrative review of applications, materials, processes, challenges, and future directions. The International Journal of Advanced Manufacturing Technology, 1-43.

[14] Cican, G., & Mitrache, A. (2025). Numerical Simulation of Paraffin Energetic Performance Enhanced by KNO3, NH4NO3, Al, Ti, and Stearic Acid for Hybrid Rocket Applications. Fuels, 6(3), 54.

[15] Koçak, D., & Özdamar, A. (2025). Numerical geometry optimization of a hybrid rocket engine using finite volume method. Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering, 09544100251328433.

[16] Rupert, K., & King, K. (2025). Effect of Oxidizer Injector Swirl Angle on Hybrid Rocket Performance with Additively-Manufactured Fuel. In 2025 Regional Student Conferences (p. 98301).

[17] Hashim, S. A., Anghan, P. P., Arah, A. A., Ahmed, M. A., Batabyal, P., Mishra, D., ... & Roy, A. (2025). Impact of zirconium nanoparticles on boron-paraffin wax-based hybrid propellant in ducted rocket applications. Engineering Research Express, 7(2), 025536.

[18] Thibaudeau, R. J., & Whitmore, S. A. (2025). A Heterogeneous Fuel Infusion Technique for Ignition and Performance Augmentation of Hybrid Rocket Engines. In AIAA SCITECH 2025 Forum (p. 2788).

[19] Bondarenko, M., Habrinets, V., & Vorobei, M. (2025). Preliminary Design Evaluation of Solid- Propellant Rocket Engines. Challenges and Issues of Modern Science, 4(2), 279-279.

[20] Arora, T. P., Buttrey, N., Kirman, P., Khadtare, S., Kamath, E., del Gatto, D., & Isoldi, A. (2025). Designing and Testing of HDPE–N2O Hybrid Rocket Engine. Engineering Proceedings, 90(1), 34.

[21] Bansal, L., & Jindal, P. (2025). Systematic evaluation of additive loading effects on burn rate, density, and mass flow in PVC/AP composite solid propellants. Aerospace Science and Technology, 110452.

[22] Dubey, A., Kumar, R., & Biswas, S. (2025). Effect of Varying Secondary Swirl Injector Location on Performance of Hybrid Rocket. Journal of Propulsion and Power, 41(6), 764-777.

[23] Srivastava, S., Thakur, A. K., Gupta, L. R., & Singh, R. (2024). Experimental test analysis of a 300 N hybrid rocket engine. Aerospace Systems, 7(4), 709-719.

[24] Ismail, I. I., Abd Razak, N., & Azami, M. H. (2024). Optimizing Nozzle Expansion Ratios with Energetic Additives for Enhanced Hybrid Rocket Motor Efficiency. Semarak International Journal of Mechanical Precision Engineering, 1(1), 20-32.

[25] Wei, S. S., Li, M. C., Lai, A., Chou, T. H., & Wu, J. S. (2024). A review of recent developments in hybrid rocket propulsion and its applications. Aerospace, 11(9), 739.

[26] Ahmad, M. T., Jagannathan, A., Abidin, R., Zhahir, A., & Saniman, M. N. F. (2024). Static Hot-Fire Testing of a Green Hybrid Rocket Engine. Journal of Aeronautics, Astronautics and Aviation, 56(1S), 405- 417.

[27] Anuar, M. N. F. M., Sujana, M. J., Ismail, I. I., & Azami, M. H. (2024). Experimental Evaluation on the Effects of Energetic Additives in the Hybrid Rocket Fuel. Journal of Transport System Engineering, 13-21.

[28] Glaser, C., Hijlkema, J., Lestrade, J. Y., & Anthoine, J. (2024). Enhancing Hybrid Rocket Engine Regression Rates: The Stepped Helix Design. Journal of Propulsion and Power, 40(5), 789-800.

[29] Navalino, R. D. A., Muda, N. R. S., Hafizah, M. A. E., & Ruyat, Y. (2024). Analysis of carbon nano particle variant as the propellant fuel to increase specific impulses of rockets. F1000Research, 12, 1414.

[30] Stella, M., Zeni, L., Nichelini, L., Bellomo, N., Pavarin, D., Migliorino, M. T., ... & Barato, F. (2024). Experimental investigation of a h2o2 hybrid rocket with different swirl injections and fuels. Applied Sciences, 14(13), 5625.

[31] Nithya Mahottamananda, S., Pal, Y., Alay Hashim, S., Dinesh, M., & Trache, D. (2024). Exploring the potential of boron carbide in enhancing energy output of solid fuels for SFRJ

and hybrid rocket propulsion systems. Propellants, Explosives, Pyrotechnics, 49(1), e202300173.

[32] Cican, G., Paraschiv, A., Buturache, A. N., Hapenciuc, A. I., Mitrache, A., & Frigioescu, T. F. (2024). Experimental research into an innovative green propellant based on paraffin–stearic acid and coal for hybrid rocket engines. Inventions, 9(2), 26.

[33] Mahottamananda, S. N., Pal, Y., Yadav, N., Trache, D., Tarchoun, A. F., Abdelaziz, A., & Pang, W. (2024). Combustion of HTPB-based solid fuels containing Viton-coated Boron for hybrid rocket applications. FirePhysChem.

[34] Kumar, G., & Kumar, R. (2024). Performance Estimation by Varying the Grain Port Alignment Position in a Hybrid Rocket Motor. Defence Science Journal, 74(2).

[35] Melis, M. (2024). Topology optimization of hybrid rocket fuels for additive manufacturing (Doctoral dissertation, Politecnico di Torino).

[36] Kumar, R. (2024). Effect of change in port shape on the performance of a hybrid rocket motor. In Proceedings of the 27th National and 5th International ISHMT-ASTFE Heat and Mass Transfer Conference December 14-17, 2023, IIT Patna, Patna-801106, Bihar, India. Begel House Inc..

[37] Gallo, G., Mungiguerra, S., Savino, R., Cardillo, D., & Battista, F. (2024). Effect of grain length on GOx-paraffin hybrid rocket engines performance and regression rate. International Journal of Heat and Mass Transfer, 220, 124978.

[38] Thibaudeau, R. (2024). Development of a Potassium Permanganate Catalyst-Infused Fuel Grain for Hydrogen Peroxide Hybrid Thruster Ignition Enhancement. In AIAA SCITECH 2024 Forum (p. 0139).

[39] Rampazzo, A., & Barato, F. (2023). Modeling and CFD simulation of regression rate in hybrid rocket motors. Fire, 6(3), 100.

[40] Nguyen, C., & Thomas, J. C. (2023). Performance of additively manufactured fuels for hybrid rockets. Aerospace, 10(6), 500.

[41] Azami, M. H., Mohamad Jamil, N. A., Mohamad Anuar, M. N. F., Nordin, N. H., & Ismail, I. I. (2023). Investigation of Hybrid Rocket Motor Regression Rate Doped with High Entropy Metallic Additives. In AIAA SCITECH 2023 Forum (p. 0891).

[42] Bycko, Z. J. (2023). Effect of additive-manufactured fuel grain geometry on small-scale hybrid rocket performance (Master's thesis, Oklahoma State University).

[43] Martins, P. G., de Souza, K. M., Boschi, R. F., Gouvêa, L. H., & Martins, C. A. (2023). Performance Comparison of Paraffin/Ethanol Fuel Blends in a Laboratory-Scale Hybrid Rocket Motor. Journal of Propulsion and Power, 39(5), 696-708.

[44] Srivastava, S., & Thakur, A. K. (2022). Review on hybrid rocket engine: past, present and future scenario. International Journal of Vehicle Structures & Systems, 14(5), 680-685.

[45] Kronwall, M. (2022). The development and implementation of a hybrid rocket motor thrust stand to investigate the relationship between combustion chamber pressure and graphite rocket nozzle erosion in hybrid rocket motors (Master's thesis, Colorado State University).

[46] Kahraman, M., Ozkol, I., & Karabeyoglu, M. A. (2022). Regression rate enhancement of hybrid rockets by introducing novel distributed tube injector. Journal of Propulsion and Power, 38(2), 200-211.

[47] Glaser, C., Hijlkema, J., & Anthoine, J. (2022). Evaluation of regression rate enhancing concepts and techniques for hybrid rocket engines. Aerotecnica Missili & Spazio, 101(3), 267-292.

[48] Tian, H., Meng, X., Zhu, H., Li, C., Yu, R., Zhang, Y., & Cai, G. (2022). Dynamic characteristics study of regression rate in variable thrust hybrid rocket motor. Acta Astronautica, 193, 221-229.

[49] Pal, Y., Mahottamananda, S. N., Palateerdham, S. K., Subha, S., & Ingenito, A. (2021). Review on the regression rate-improvement techniques and mechanical performance of hybrid rocket fuels. FirePhysChem, 1(4), 272-282.

[50] Gieras, M., & Gorgeri, A. (2021). Numerical modelling of the hybrid rocket engine performance. Propulsion and Power Research, 10(1), 15-22.

[51] Barato, F., Toson, E., & Pavarin, D. (2021). Variations and control of thrust and mixture ratio in hybrid rocket motors. Advances in Astronautics Science and Technology, 4(1), 55-76.

[52] Gallo, G., Mungiguerra, S., & Savino, R. (2021). New entrainment model for modelling the regression rate in hybrid rocket engines. Journal of Propulsion and Power, 37(6), 893-909.

[53] Thomas, J. C., Paravan, C., Stahl, J. M., Tykol, A. J., Rodriguez, F. A., Galfetti, L., & Petersen, E. L. (2021). Experimental evaluation of HTPB/paraffin fuel blends for hybrid rocket applications. Combustion and Flame, 229, 111386.

[54] Madhusudhan, N. K., & Micklow, G. (2020). Numerical and Thermochemical Analysis of Hybrid Rocket Fuels with Metallic Additives. Multidisciplinary Engineering Science and Technology, 7(11).

[55] Cai, G., Zhao, Z., Zhao, B., Liu, Y., & Yu, N. (2020). Regression rate and combustion performance investigation on hybrid rocket motor with head-end swirl injection under high geometric swirl number. Aerospace Science and Technology, 103, 105922.

[56] Pal, Y., Raja, A., & Gopalakrishnan, K. (2020). Theoretical and experimental heat of combustion analysis of paraffin-based fuels as preburn characterization for hybrid rocket. Journal of Aerospace Technology and Management, 12, e4520.

[57] Elanjickal, S., & Gany, A. (2020). Enhancement of the fuel regression rate in hybrid propulsion by expandable graphite additive. Combustion Science and Technology, 192(7), 1253-1273.

[58] Ferrero[1], A., Masseni, F., Muscarà, L., & Pastrone, D. (2020). Multiphysics Modelling of a Hybrid Rocket Engine.

[59] Ozawa, K., & Shimada, T. (2020). Performance of mixture-ratio-controlled hybrid rockets for nominal fuel regression. Journal of Propulsion and Power, 36(3), 400-414.

[60] Teja, P. S., Sudhakar, B., Dhass, A. D., Krishna, R., & Sreenivasan, M. (2020). Numerical and experimental analysis of hydroxyl-terminated poly-butadiene solid rocket motor by using ANSYS. Materials Today: Proceedings, 33, 308-314.e studies.